DHA Suffa University

Final Year Project

Emotion Aware Emergency Dispatch System (EAEDS)

S31F25

Software Design Specifications

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Definition of Terms, Acronyms, and Abbreviations

Terms	Description
EAEDS	Emotion Aware Emergency Dispatch System, which is a system designed to analyze the emotions of callers and help dispatchers in their work.
LLM	A model that is train on alot of data like whole world data like chat gpt,gemini and many more.
Whisper	an open-source ASR model created by OpenAI that is used for converting speech into text.
Latency	the time it takes for a system to respond to an action.
GPU	Graphics Processing Unit, specialized hardware that speeds up AI calculations.
FastAPI	Python framework used in our project backend for creating apis.
Next.js	React framework used for frontend in our project.
SQLite	Database for storing calls and other works
Dashboard	the main page where live call information, emotions, and location of caller is displayed. In our project it is called midnight dashboard
JSON	JavaScript Object Notation. A structured data format used to store transcripts and emotion data.
UI/UX	Design of how our dashboard looks
SQLite	Database for storing calls and other works
Call Session	A single emergency interaction that includes audio, transcript, emotion data, and metadata such as time and location.
BERT	Bidirectional Encoder Representations a model used for understanding natural language and analyzing emotions.
Whisper	an open-source ASR model created by OpenAI that is used for converting speech into text.
Open Smile	an open-source tool that extracts acoustic and emotional features from audio signals.

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Introduction

Purpose of Document

This document perform its function as the blueprint for the design and for the application of the Emotion Aware Emergency Dispatch System(EAEDS). It is basically designed for the developing the team, project supervisors, and stakeholders and they all will be included in reviewing, performing , and using the system. Object-Oriented Design (OOD) is employed as the design methodology in this project and it matches with the structured and recyclable nature of the software components.

Intended Audience

The intended audience for this document includes:

- **Project Team**: Programmers and verifiers are responsible for constructing and checking the system.
- **Supervisors**: The project's progress is supervised by the academic advisors.

Document Convention

- **Font:** TIMES NEW ROMAN
- **Font Size:** 12pt for body text, 14pt for headings
- **Formatting:** Consistent use of bullet points, numbered lists, and bold headings for clarity.

Project Overview

The conventional 911 response system is usually improved bu the EAEDS which is commonly also called as DispatchAI. It helps in handling the worldwide problem of call centers which is understaffed by 82% by adding up an AI assistant which performs its work alongside human programmers. The system can listen, understand, and think with the programmers . It works by using tools like Whisper, OpenSMILE, BERT, and a fine-tuned language model to change audio into applicable data, which includes transcripts, sentiment analysis, and location information.

Scope

What the System Will Do:

This project's range includes the design and creation of functional prototype for an Autonomous Dispatch Assistant. The main features of this include:

Real-time Speech-to-text: Instantly changes caller speech into text using Whisper model.

Emotion Detection: Inspecting vocal patterns with OpenSMILE to detect emotional states such as panic, fear, or neutrality.

Smart Decision-Making: Using fine-tuned LLMs to follow standard 911 processes (Medical, Fire, Police) and produce questions and suggestion according to current scenario of caller.

Operator Dashboard: A web-based interface (built with Next.js) that shows live transcripts, detected emotions, and recommended actions for dispatchers and call history.

Local GPU Architecture: A unified system where all AI processing and user interface components run on the local system equipped with NVIDIA GPU hardware, eliminating cloud dependencies and reducing latency..

What the System Will Not Do:

The following items are clearly removed from this phase of the project:

PSTN/Telephony Integration: Currently, the system uses calls using WebSockets and does not connect to original landline or mobile networks.

Legal/Medical Liability: The system only provides recommendations and is not meant to make final legal or medical decisions without a human checking them.

Physical Hardware Deployment: This project is software based only and does not cover giving physical servers or specialized radio equipment.

Multi-Language Support: Our project only works with english language not with other languages so it is not multi language project.

Design Considerations

Data Collection and Preparation:

High-fidelity real-time audio streams is controlled by the system. Audio is recorded through WebSockets, and it was similar to 16kHz mono, and stored temporarily to make sure alignment with Whisper (ASR) and OpenSMILE (Acoustic Emotion) models. Transcription data is made safe and organised before it is analysed by the LLM for response generation.

ML Model Integrated Design:

The architecture combines many professional models:

Whisper: Provide accurate speech to text.

BERT + OpenSMILE: It is used for both meaning-based and sound-based emotion classification.

Llama 3/Mistral: It is used for advanced and responsive generation.

The design focus on low lag and understandability , and present emotion measures and provide suggestions to dispatchers in a Human in the loop format.

Reliability and Scalability:

The system uses a

UI/UX Strategy:

The Midnight Dashboard is formed for handling high-stress environments. It provieds a dark mode tactical interface along with the real-time transcript scrolling and dynamic mapping through`Leaflet.js`, and easy to use emotion alerts which decrease dispatcher cognitive load.

Maintenance and Evolution:

The modular design give permission for hot swapping models. The AI service interface make sures that updating from BERT to RoBERTa or shifting LLMs needs less code changes.

Assumptions:

Technical Dependencies:

- **FastAPI:** used for backend and apis.
- **Next.js:** used for main dashbaord devlopment mean frontend
- **SQLite:** Lightweight,efficient database used for call history
- **Local GPU:** On-premises GPU (minimum GTX 1660 or higher) for running AI models

- Audio input is considered to be in English for the current model weights.
- The user contains a browser that is capable to support WebSockets and Leaflet rendering.

Risks and Volatile Areas:

Latency Risks:

Processing large audio chunks on local GPU may cause latency if hardware is insufficient. Contingency: Implementation of batch processing optimization, model quantization, and hardware requirement validation to ensure response time under 2 seconds.

Model Accuracy: It is the detection of emotions during high stress, noisy environments might be sensitive .

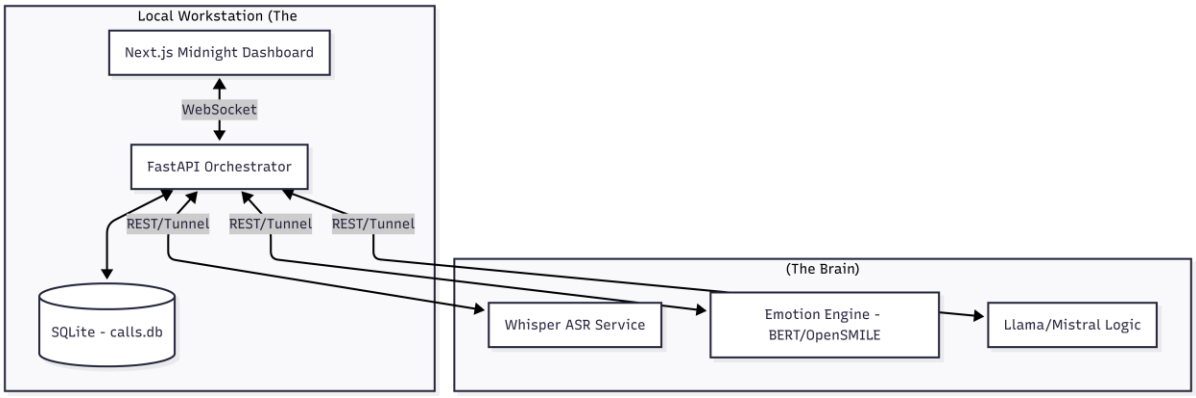
- **Contingency:** Using a double engine approach like Acoustic along with Semantic for verification of indicators.

Scaling Volatility:

Handling multiple simultaneous high bitrate audio streams might strain local GPU VRAM and system memory. Contingency: Implementation of queue-based processing and GPU memory management with maximum concurrent call limits based on available VRAM.

- **Contingency:** Usage of local streaming and horizontal scaling of AI worker nodes.

System Architecture :



High-Level Decomposition:

The system is divided in three large subsystems:

Frontend Dashboard: It is the real time visualization and user interaction.

Backend Orchestrator: It is the routing , management of session and data persistence.

AI Engine: It is heavy audio working through computation and reasoning.

Physical Design:

Execution: The entire system including Frontend, Backend, and AI Engine runs on a single local workstation (Windows/Linux) equipped with NVIDIA GPU.

GPU Layer: All AI models (Whisper, BERT, OpenSMILE, LLM) execute on the local NVIDIA GPU (minimum 6GB VRAM recommended, 8GB+ for optimal performance).

Error Handling: Disconnection are handled by complete try except blocks in the WebSocket loop and partial transcripts are saved by complete try except blocks in the WebSocket loop which is done to prevent data loss.

Software Architecture :

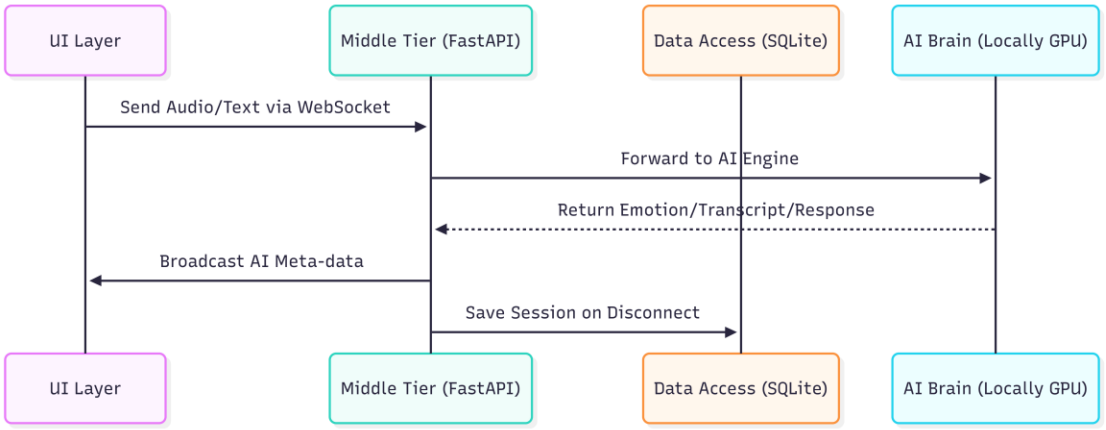
The system comes after a simple Three-Tier Architecture:

User Interface Layer: It is mainly constructed with Next.js, providing the "Phone UI" for callers and "Midnight Dashboard" for users.

Middle Tier (Business Logic): FastAPI works by controlling WebSocket bi-directional streaming, orchestrating AI service calls, and controlling local business rules.

Data Access Layer: SQLite operations are abstracted by database manager class for protecting and retrieving call logs.

GPU Resource Management Layer: PyTorch/CUDA interface manages model loading, inference batching, and VRAM allocation on the local GPU to ensure optimal performance across all AI components.

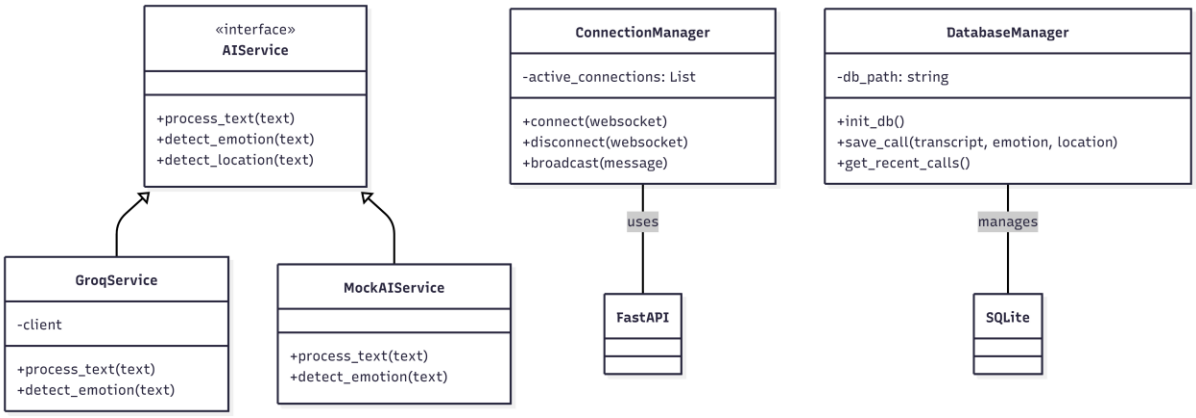


Design Strategy:

- **Future Extension:** It works by using an Abstract Base Class AIService which helps to allow simple integration of new LLMs or ASR models.
- **System Reuse:** Components like ConnectionManager are constructed for any real-time broadcasting requirement .
- **Data Management:** It works by using JSON strings in SQLite to save complex transcript structures.
- **Concurrency:** It uses Python's async/await and FastAPI's Websocket abilities to control simultaneous connections without any delay.

Detailed System Design

Design Class Diagram



Er Diagram(Single-Entity Model)

CALL		
string	id	PK
datetime	started_at	
datetime	ended_at	
string	caller_location	
string	detected_emotion	
string	transcript_json	
int	duration_seconds	

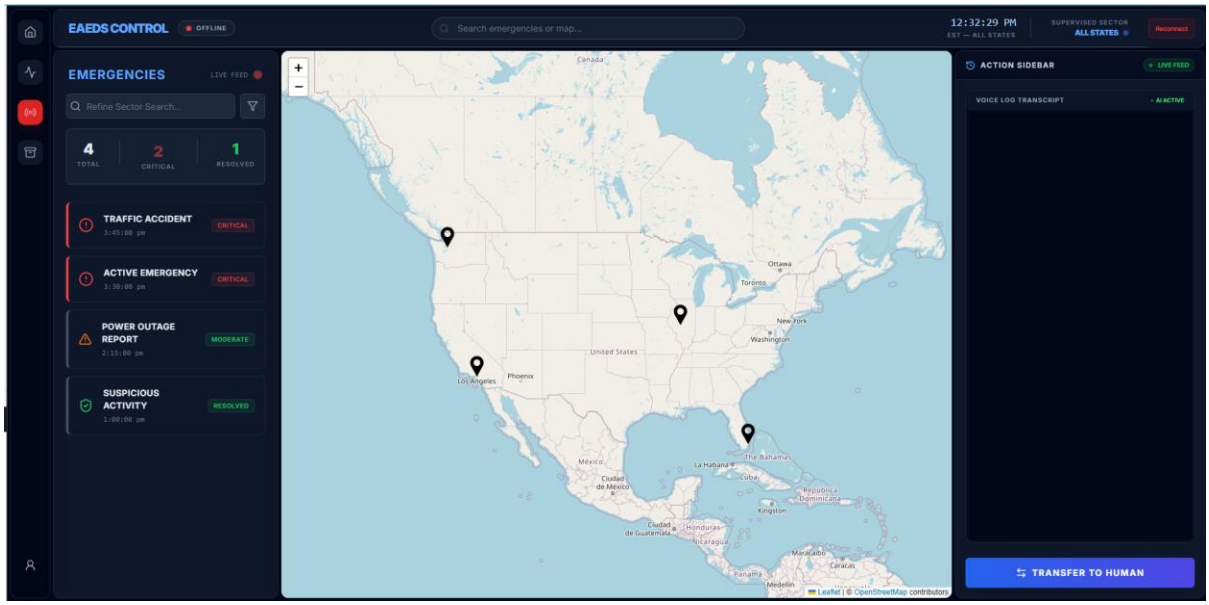
The `CALL` entity stores all metadata for a single calling session. It includes time, location, emotional state, and whole talking which is stored as json format.

Data Dictionary

< Data 1>						
Name	Call_Record					
Alias	Emergency_Session, Call_Log					
Where-used/how-used	DatabaseManager, History Dashboard, AI Analytics`					
Content description	Composition of a single emergency interaction.					
Column Name	**Description**	**Type**	**Length**	**Null able**	**Default Value**	**Key Type**
id	Global Unique ID	text	36	no	UUIDv4	PK
started_at	Session Start Time	datetime	-	no	CURRENT_TIMESTAMP	-
ended_at	Session end time	datetime	-	yes	-	-
caller_location	Address	text	255	yes	Unknown	-
detected_emotion	Final emotion score	text	50	no	neutral	-
transcript	full json call	text	-	no	[]	-
duration	call length in seconds	integer	-	no	0	-

GUI:

Mock Screen 1:



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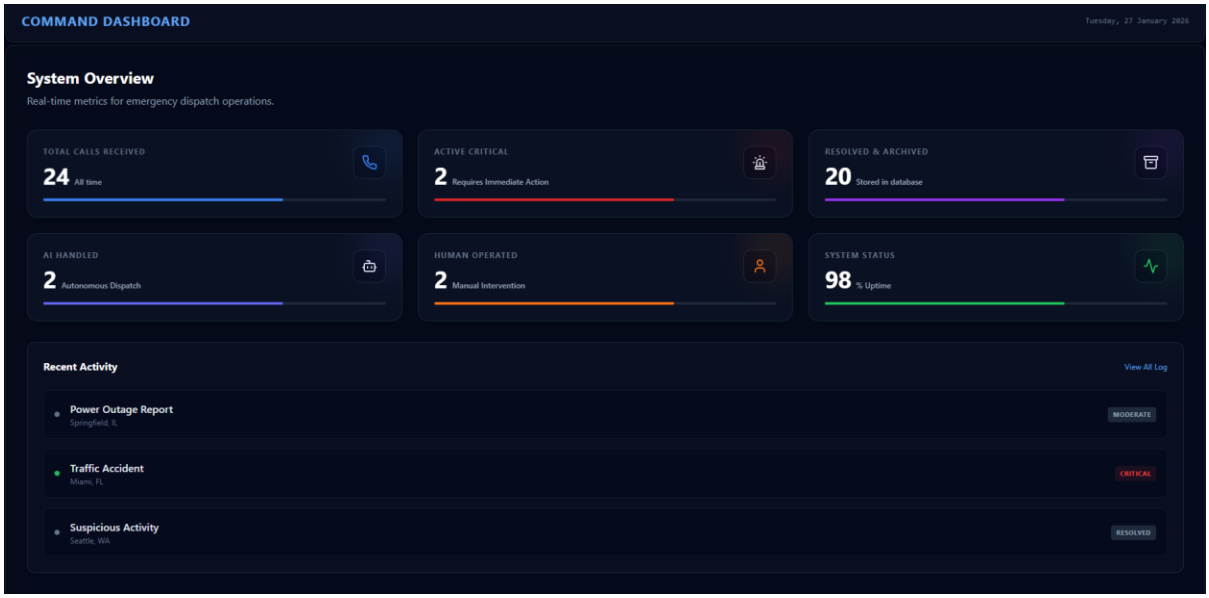
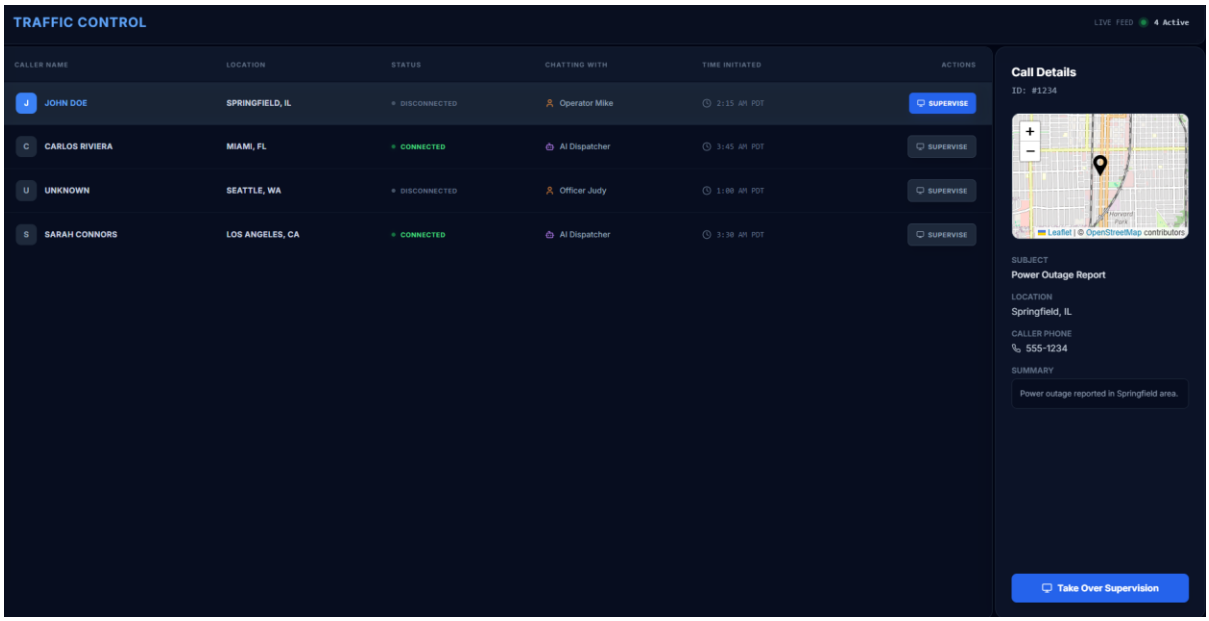
Mock Screen 3: